

probability each
point gets for
its **true** class

0.7, 0.4, 0.4, 0.8

multiply: likelihood
(maximise)

$$0.7 \times 0.4 \times 0.4 \times 0.8 = 0.090$$

take logs: product
becomes a sum

$$\log 0.7 + \log 0.4 + \dots = -2.41$$

change sign:
cross entropy
(minimise)

2.41